

Computer vision - Assignment 2

1. Download the images 'image1' and 'image2' from the Google folder labeled '*Histogram assignment*'.
 - (a) Extract the 3 color channels for each image and plot their respective histograms (so six in all).
 - (b) For each channel, match the histograms of each colour channel of image1 with corresponding channel histograms of image2. Use `cv2.equalizeHist()`.
 - (c) Reconstruct the new image1 in colour.
 - (d) Compare the result with the original images.
2. Pick a grayscale image, either from the given ones or from scikit's image repository. Perform intensity slicing on the image in two ways:
 - (a) highlight a particular range and set the rest to zero;
 - (b) highlight a particular range and leave the rest unchanged.

In each case, plot the intensity transformation being applied. Display all results for comparison.

3. Perform bit-plane slicing on the same grayscale image used in Exercise 2.
 - (a) Display all the bit planes, labeling them as 'bitplane0'(corresponding to the least significant bit) to 'bitplane7'(corresponding to the most significant bit).
 - (b) Plot the intensity transformation being used to extract bitplane0, bitplane3 and bitplane7.
 - (c) Try to reconstruct the image with fewer bit planes so as to visually appear as resembling the original. What is the minimum number of bit planes the you can use without degrading the image quality too much?
4. Aligning the 3 channels of the Prokudin-Gorskii Photo Collection:

For this assignment we have chosen one of the images in the above mentioned collection - windmills.jpg. Download this image, split the image into three parts to extract the channels. The first channel is the blue channel, followed by the green channel, and the red channel. Keep the blue channel fixed and align the green and red channels to the blue channel in the following way:

- (a) To compute the best alignment between two channels, "slide" one image over the other i.e. search over a window of possible displacements say $[-10, 10]$ pixels, score each one using an image matching metric such as SSD, and take the displacement with the best score.
- (b) What is your best SSD value and displacement vector for each channel?
- (c) Once you find the best alignment between the channels, merge the 3 channels to get a coloured image.
- (d) Display the coloured image.